

Debosmita (Deb) Pathak

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Education

Ph.D., Astronomy , The Ohio State University, Columbus, OH	2022 – Present
M.S., Astronomy , The Ohio State University, Columbus, OH	2022 – 2024
B.A. with Honors, Physics & Mathematics , Grinnell College, Grinnell, IA	2018 – 2022

Research Experience

IPAC Visiting Graduate Student Research Fellow	Pasadena, CA
<i>Infrared Processing and Analysis Center, California Institute of Technology</i>	Feb 2026 – Aug 2026
Advisors: Dr. Lee Armus, Dr. Laura Lenkic, Dr. Thomas Lai, Dr. Vivian U	

- Measured stellar feedback pressures and the dynamical state of young star clusters in the closest Luminous Infrared Galaxy, NGC3256, from the Great Observatories All-sky LIRG Survey (GOALS), using JWST, VLT/MUSE, ALMA, and HST data.

Visiting Scholar	San Diego, CA
<i>University of California, San Diego</i>	Sept 2024 – Dec 2024
Advisor: Dr. Karin M. Sandstrom	

- Characterized the variations in mid-infrared polycyclic aromatic hydrocarbon (PAH) emission in the diffuse interstellar medium of 71 nearby star-forming galaxies using JWST/MIRI, JWST/NIRCam, VLT/MUSE, ALMA, and *Herschel*/PACS data.

Graduate Fellow	Columbus, OH
<i>The Ohio State University</i>	Aug 2022 – Present
Advisors: Dr. Adam K. Leroy, Dr. Todd A. Thompson, Dr. Laura A. Lopez	

- Improving optical-based predictions for cluster mass due to stochastic IMF sampling using SLUG cluster simulations and multi-wavelength photometry for 18,000 PHANGS HII regions.
- Measured the mass components, star-formation efficiencies, and pressure-balance in 18,000 HII regions across 19 star-forming galaxies using JWST/NIRCam, HST, VLT/MUSE, ALMA, VLA, and MeerKAT.
- Estimated stellar feedback pressures, including radiation and warm gas pressure, in 18,000 HII regions across 19 nearby star-forming galaxies using JWST/MIRI, HST, and VLT/MUSE.
- Measured the probability distribution functions of dust emission and total gas column density in the disks of 19 nearby star-forming galaxies using JWST/MIRI and VLT/MUSE.

Undergraduate Researcher	Grinnell, IA
<i>Grinnell College</i>	2021 – 2022
Advisor: Dr. Charlotte R. Christensen	

- Compared the timescales of quenching and disruption in satellite galaxies around simulated Milky Way analogs using high-resolution cosmological zoom-in simulations.

Undergraduate Researcher	(virtual) Boston, MA
<i>Harvard-Smithsonian Center for Astrophysics</i>	2020 – 2021
Advisors: Dr. Sirio Belli, Dr. Rainer Weinberger	

- Studied observable signatures of major merger-driven starbursts and rapid-quenching in high-redshift post-starburst galaxies using IllustrisTNG cosmological hydrodynamical simulations.

Publications

6 as first author, 18 as contributing author [↗](#)

1. **Pathak, D.** et al. (2026). Mid-Infrared Colors Vary with Galactic Environment: Contrasting Star-Forming Disks, Young Centers, and Quiescent Star-Formation Deserts (*ApJ*)
2. **Pathak, D.** et al. (2025). Masses, Star-Formation Efficiencies, and Dynamical Evolution of 18,000 HII Regions (*ApJL*)

3. **Pathak, D.** et al. (2025). [Survivors and Zombies: The Quenching and Disruption of Satellites Around Milky Way Analogs](#) (*ApJ*)
4. **Pathak, D.** et al. (2025). [Linking Stellar Populations to HII Regions across Nearby Galaxies. II. Infrared Reprocessed and UV Direct Radiation Pressure in HII Regions](#) (*ApJ*)
5. **Pathak, D.** et al. (2024). [A Two-Component Probability Distribution Function Describes the Mid-IR Emission from the Disks of Star-forming Galaxies](#) (*AJ*)
6. **Pathak, D.** et al. (2021). [Quenching, Mergers, and Age Profiles for \$z = 2\$ Galaxies in IllustrisTNG](#) (*ApJL*)

Honors & Awards

Allan H. Markowitz Award in Observational Astronomy, OSU	2025
AAS International Travel Grant, IAU General Assembly	2024
AAS Chambliss Honorable Mention	2024
NSF Graduate Research Fellowship, OSU	2022-Present
Susan Huntington Dean's Distinguished University Fellowship, OSU	2022-Present
H. George Apostle Prize in Physics, Grinnell College	2022
Leopold Schepp Foundation Scholarship, Grinnell College	2021-2022
AAS Chambliss Honorable Mention	2021

Software

Languages: Python, R, SQL, C++, C, JavaScript

Astronomy, Image Processing, and Machine Learning: astropy, scikit-image, SLUG, pyneb, OpenCV, pandas, SciPy, DS9, CARTA, scikit-learn, UMAP